ESTION 1: [6 POINTS]

ect the correct answer.

A. **[1 POINT]** The procedural/structured paradigm focuses on functions, whereas the object-oriented paradigm combines data and methods into objects.

a. True

b. False

B. **[1 POINT]** nextChar() method in the Scanner class is used to read a character in Java.

a. True

b. False

C. **[1 POINT]** In Java, a constructor must always have the same name as the class and can return a value.

a. True

b. False

D. **[1 POINT]** The _____ access modifier makes members accessible from everywhere within the java project.

a. static

b. private

c. public

d. Default

E. **[1 POINT]** The process in Java that allows defining two or more methods within the same class that share the same name, as long as their parameters are different, is:

a. Inheritance.

b. Encapsulation.

c. Overloading.

d. Polymorphism.

F. **[1 POINT]** Which keyword is used to define a **constant** in Java ?

a. final

b. static

c. constant

d. private

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ESTION 2: Evaluate the following Java codes and show the exact output. **[4.5 POINTS]**

A. [2 points]

```
public class NewClass {
    static int Count = 0;
    private int x;
    public NewClass() {
        this(1);
        ++Count;
    }
    public NewClass(int i) {
        x = i;
        ++Count;
    }
    public String toString() {
        return x + " " + Count;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        System.out.println(NewClass.Count);
        NewClass q1 = new NewClass();
        NewClass q3;
        System.out.println(NewClass.Count);
        q3 = q1;
        System.out.println(q3);
    }
}
```

B. [1 point]

```
public static void main(String[] args) {
   double x=1+Math.random()*10;
        if(x<11)
            System.out.println("welcome to java");
        else
            System.out.println("I love java");
}
```

welcome to Java

C. [1.5 points]

```
public static void main(String[] args) {
        String s1="Hello ";
        String s2="oops";
        String s3=s1.charAt(4)+s2.substring(1,3)+" 1";
        System.out.println(s3);
        String s4=s1.concat(s3);
        System.out.println(s4);
        System.out.println(s4.indexOf('h'));
    }
```

OOP 1

Hello OOP 1

-1

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ESTION 3: Analyze each line of the following code and write in the table below whether the cified lines have an error or not. If the line has an error, then explain why. **[5.5 POINTS]**

```
public class Vehicle {
  private int wheels;
  private int doors;
  private int seats;
  public Vehicle() {
    this.wheels = 4;  ----------------------------------------------------------> //line 1

    doors = 4;
  }

  public static Vehicle(int wheels, int doors, int seats) { ----------------> //line 2
    this.wheels = wheels;
    this.doors = doors;
    this.seats = seats;
  }

  public static boolean canCarry(int numberOfPeople) {
    return (seats >= numberOfPeople); ---------------------------------------> //line 3
  }
}
```

```
public class TestVehicle {
  public static void main(String[] args) {
    Vehicle car1 = new Vehicle(); ---------------------------------------------> //Line 4

    Vehicle car2 = new Vehicle(4); --------------------------------------------> //Line 5

    Vehicle car3 = new Vehicle(4, 2, 5);

    car1.doors = 2; -----------------------------------------------------------> //Line 6

    System.out.println(car1); -------------------------------------------------> //Line 7
  }
}
```

3.5

Line	Correct/Incorrect	Explanation why if incorrect
Line 1	~~incorrect~~	
Line 2	~~incorrect~~	constructor should ~~not be~~ static
Line 3	~~incorrect~~	~~seats is a private i seats is not declare~~
Line 4	~~correct~~	
Line 5	~~correct~~	
Line 6	~~incorrect~~	it's ~~private~~ should be accessed with set
Line 7	~~correct~~	

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composition 13.5

QUESTION 4: [14 POINTS]

Apply object-oriented programming principles to develop a solution for the given problem:

A. [5.25 POINTS] Write a java class named **QuestionClass**. This class contains the following:

Assignment has a question has-a

- A private instance variable: **questionText** of type String, which represents the text of the question.
- A parameterized constructor with one parameter to initialize **questionText**.
- A public method **getQuestionText** to return the instance variable of the class.
- A public method **setQuestionText** that assigns text to the question. This method includes validation to ensure that the text is not null. If the text is found to be null, the method will store **"No question provided"**.

B. [5.25 POINTS] Write a java class named **Assignment** that contains the following:

- Two instance variables:
  - A private **assignmentTitle:** of type String.
  - A private **question**: This instance variable should be a composition of the **QuestionClass**. object
- A parameterized constructor with two parameters to initialize **assignmentTitle** and **questionText** in **question**.
- A **toString()** method that returns the details of the **assignmentTitle** and **questionText** from the **question**.

C. [3.5 POINTS] Create a Java class called **Test** and write a main method that performs the following actions:

- Ask the user to enter details of the Assignment, including the **assignmentTitle** and **questionText**.
- Declare an object of **Assignment** type with the values entered by the user: **assignmentTitle** and **questionText**.
- Print all the information of the Assignment object.

```
public class QuestionClass()
{ private String questionText;
  public QuestionClass(string QT)
  {
    setquestionText(QT);
  }
  public string getquestionText()
  { return questionText; }
```

0.25 0.5 0.75 2.5

```
Public void setquestionText(String question)
{
  if(question != "null")
    questionText = question
  else
    systemout.Println("No question provid
  }
}
```

1

```
public class Assignment ( )        0.25
{
  private String AssignmentTitle ;   0.5
  private QuestionText question ;    0.75

 public Assignment (String AT , string questionText )
 {
  AssignmentTitle = AT ;   2
  Question class (questionText) ;   2

 }
 public String toString ( )
 {
  return " AssignmentTitle : " + AssignmentTitle + " /n " + " questionText " +
                                                          question ;
  }
                1.25
 }
```

```
import java.util.Scanner();

Public class Test( )
                0.25
{
public static void ~~to~~ main 0.25 (String [ ] args)
{
  system.out.println(" enter assignment title")

0.25 scamer read = new Scanner(system.in);

    string x = ~~text~~ read.nextLine( ); 0.5

      s.o.p(" enter question text" );

      String y = read.nextLine( ) ; 0.5

      Assignment obj = new Assignment(x , y);
1.25

0.5   s.o.p(obj);
```

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